WORKED SOLUTION

STEP II 2001, Question 11

Mechanics • Projectile motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

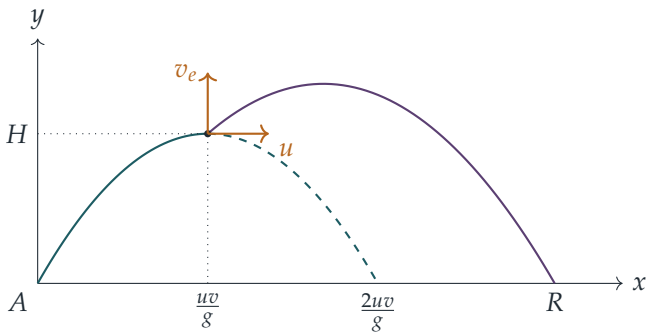
step.maths.org/questions/01-s2-q11

The key idea

Split the flight at the explosion. Before it, we have an ordinary projectile, and the highest point is where the vertical velocity is zero. After it, fragment P is a new projectile launched from that height with velocity (u, v_e), so the time to land comes from a quadratic. For the second part, the range is an increasing function of v_e, so "$R > D$" turns into "v_e exceeds the value that gives $R = D$ exactly".

Solution

We model the missile and fragment P as particles moving under gravity alone, with the ground horizontal. Take A as origin, x horizontal and y vertically upwards.



The dashed curve is where the missile would have landed with no explosion; the purple curve is the path of P.

The range of P

Up to the explosion. The vertical velocity is $v - gt$, which vanishes at $t_1 = v/g$. At that moment the missile has travelled horizontally

$$x_1 = ut_1 = \frac{uv}{g},$$

and its height, from $v^2 = 2gH$ with final vertical speed zero, is

$$H = \frac{v^2}{2g}.$$

After the explosion. Fragment P starts at height H with horizontal velocity u and vertical velocity v_e upwards. Measuring time t from the explosion, its height is

$$y = H + v_e t - \tfrac{1}{2}gt^2.$$

It lands when $y = 0$:

$$\tfrac{1}{2}gt^2 - v_e t - \frac{v^2}{2g} = 0 \quad \Longleftrightarrow \quad g^2t^2 - 2gv_e t - v^2 = 0.$$

Treating this as a quadratic in gt,

$$gt = v_e \pm \sqrt{v_e^2 + v^2}.$$

The product of the roots is $-v^2 < 0$, so one root is negative. That root is the time *before* the explosion at which the parabola would cross the ground, and it is irrelevant. So the flight time after the explosion is

$$t_2 = \frac{v_e + \sqrt{v_e^2 + v^2}}{g}.$$

The horizontal velocity stays u throughout, so the total horizontal distance is

$$R = \frac{uv}{g} + ut_2 = \frac{u}{g}\left(v + v_e + \sqrt{v_e^2 + v^2}\right),$$

that is,

$$\frac{gR}{u} = v + v_e + \sqrt{v_e^2 + v^2},$$

as required.

The condition for $R > D$

Write

$$f(v_e) = v_e + \sqrt{v_e^2 + v^2}, \qquad k = \frac{gD}{u} - v = \frac{gD - uv}{u}.$$

Then $R > D$ is equivalent to $f(v_e) > k$. Since $D > 2uv/g$, we have $gD > 2uv$, so $k > v > 0$.

Ask yourself Why is it enough to find the single value of v_e that makes $R = D$?

Because f is strictly increasing:

$$f'(v_e) = 1 + \frac{v_e}{\sqrt{v_e^2 + v^2}} > 0,$$

since $|v_e| < \sqrt{v_e^2 + v^2}$ when $v \neq 0$. (Physically, a harder upward kick keeps P in the air longer.) So if $f(v_e^*) = k$, then $f(v_e) > k$ exactly when $v_e > v_e^*$.

To find v_e^*, solve $\sqrt{v_e^2 + v^2} = k - v_e$. Squaring gives

$$v_e^2 + v^2 = k^2 - 2kv_e + v_e^2 \quad \Longrightarrow \quad v_e^* = \frac{k^2 - v^2}{2k} = \frac{(k-v)(k+v)}{2k}.$$

Squaring could introduce a false root, so we check that $k - v_e^* \geq 0$: indeed $k - v_e^* = \dfrac{k^2 + v^2}{2k} > 0$, so v_e^* really does satisfy $f(v_e^*) = k$.

Now substitute for k:

$$k - v = \frac{gD - 2uv}{u}, \qquad k + v = \frac{gD}{u}, \qquad 2k = \frac{2(gD - uv)}{u},$$

so

$$v_e^* = \frac{(gD - 2uv)\, gD/u^2}{2(gD - uv)/u} = \frac{gD}{2u}\left(\frac{gD - 2uv}{gD - uv}\right).$$

Hence, since f is increasing, $R > D$ whenever

$$v_e > \frac{gD}{2u}\left(\frac{gD - 2uv}{gD - uv}\right),$$

which is the required condition. In fact the argument shows it is also necessary. Note that $v_e^* > 0$ because $k > v$, so a genuine upward kick is needed.

Checks

- If $v_e = 0$ the formula gives $gR/u = 2v$, so $R = 2uv/g$: the usual range of an unexploded projectile, as it should be. This is also why the question assumes $D > 2uv/g$, since otherwise no kick is needed.
- With $u = v = 10$, $g = 10$ and $D = 30$, the threshold is $v_e^* = \frac{300}{20} \cdot \frac{100}{200} = 7.5$, and then $gR/u = 10 + 7.5 + 12.5 = 30$, so $R = 30 = D$ exactly.
- A numerical step-by-step simulation of the two-stage flight agreed with the range formula to about 10^{-9}, and 10^5 random cases confirmed that $R > D$ exactly when $v_e > v_e^*$.

Common slips

- Forgetting the horizontal distance uv/g covered before the explosion, and measuring R from the point below the explosion instead of from A.

- Taking the time to land from the top as v/g again, which ignores the new upward velocity v_e.
- Squaring $\sqrt{v_e^2 + v^2} = k - v_e$ without checking that the right-hand side is non-negative, or without saying why one equality value turns into an inequality.

Beyond STEP

The explosion conserves the momentum of the whole system, so the centre of mass of the two fragments carries on along the original parabola even though P does not. That principle, that internal forces cannot change the motion of the centre of mass, is the starting point for rocket motion and collisions in a first-year mechanics course. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both carefully.

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